

Exercise for Sect. 2.9

Exercise

2.9.1

Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system, and let $r : X \rightarrow \mathbb{N}_0$ be a map in

L^1_μ . The suspension defined by r is the system $(X^{(r)}, \mathcal{B}^{(r)}, \mu^{(r)}, T^{(r)})$, where

- $X^{(r)} = \{(x, n) : 0 \leq n < r(x)\}$;
- $\mathcal{B}^{(r)}$ is the product σ -algebra of $\mathcal{B}$ and the Borel σ -algebra on $\mathbb{N}$ (which comprise all subsets);
- $\mu^{(r)}$ is defined by $\mu^{(r)}(X \times N) = \frac{1}{\int r d\mu} \mu(A) \times |N|$ for $A \in \mathcal{B}$ and $N \subseteq \mathbb{N}$;

and

$$T^{(r)}(x, n) = \begin{cases} (x, n+1) & \text{if } n+1 < r(x) \\ (T(x), 0) & \text{if } n+1 = r(x). \end{cases}$$

1. Verify that this defines a finite measure-preserving system.
2. Show that the induced map on the set $A = \{(x, 0) : x \in X\}$ is isomorphic to the original system $(X, \mathcal{B}, \mu, T)$.

Proof:

1. Suppose $E \in \mathcal{B}^{(r)}$, we can divide E into disjoint sets $E = \bigsqcup_{n \in \mathbb{N}} E_n$ where $E_n = F_n \times \{n\}$ for some $F_n \in \mathcal{B}$. If $n \geq 1$, $\mu^{(r)}(E_n) = \mu^{(r)}(F_n \times \{n-1\}) = \mu^{(r)}(E_{n-1})$. And if $n = 0$, $\mu^{(r)}(E_n) = \mu^{(r)}(\{(y, r(y)-1) : Ty = x \text{ for some } x \in F_n\}) = \frac{1}{\int r d\mu} \mu(T^{-1}F_n) = \mu(E_n)$. Thus, $\mu^{(r)}$ is measure-preserving.

The space $X^{(r)}$ also can be divide into disjoint sets $X_n = \{(x, n) : x \in X, r(x) > n\}$. Then, $\mu^{(r)}(X^{(r)}) = \mu^{(r)}(\bigsqcup X_n) = \frac{1}{\int r d\mu} \mu(\{x \in X : r(x) > 0\}) = 1$. Thus, $(X^{(r)}, \mathcal{B}^{(r)}, \mu^{(r)}, T^{(r)})$ is a finite measure-preserving system.

2. By the definition, we can easily see that $T_A(x, 0) = (Tx, 0)$, and the bimeasurable bijection from induced map to original system is the projection $\pi : A \rightarrow X$ by $\pi(x, 0) = x$.

□

Exercise

2.9.2

An invertible measure-preserving system $(X, \mathcal{B}, \mu, T)$ is called aperiodic if $\mu(\{x \in X : T^k(x) = x\}) = 0$ for all $k \in \mathbb{Z} \setminus \{0\}$.

1. Show that an ergodic transformation on a non-atomic space is aperiodic.
2. Find an example of an aperiodic transformation on a non-atomic space that is not ergodic.
3. Prove Kautani-Rokhlin lemma for an invertible aperiodic transformation on a non-atomic space.

Proof:

□

Exercise

2.9.3

Show that for an ergodic measure-preserving system $(X, \mathcal{B}, \mu, T)$, sequence $a_1, \dots, a_n$ of distinct integers, and $\varepsilon > 0$ it is not always possible to find a measurable set A with the properties that $T^{a_1}(A), \dots, T^{a_n}(A)$ are disjoint and $\mu\left(\bigcup_{i=1}^n T^{a_i}(A)\right) > \varepsilon$.

Proof:

□

Exercise**2.9.4**

Let $(X, \mathcal{B}, \mu, T)$ be an invertible aperiodic measure-preserving system on a non-atomic space. Then, for any $\varepsilon > 0$, there is a set $A \in \mathcal{B}$ with $\mu(A) < \varepsilon$ with the property that for any finite set $F \subseteq X$, there is some $j = j(F)$ with $F \subseteq T^{-j}(A)$.

Proof:

□